



# Takoua Khammassi

Software Engineering Student — Data Science & AI

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 <https://my-portfolio-takoua.vercel.app/> Tunis, Tunisia

## PROFILE

Final-year software engineering student at ISAMM, specializing in Data Science and AI through internships and personal projects in Machine Learning, Deep Learning, and Computer Vision. I am currently seeking a final-year internship (PFE) to contribute to innovative projects in this field, while further developing my technical skills and thriving in a stimulating international environment.

## TECHNICAL SKILLS

**Programming Languages :** Python, SQL, JavaScript

**Data Science & AI :** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Machine Learning, Deep Learning, Transfer Learning, Transformers, Computer Vision, NLP, LLMs, RAG

**Big Data :** Hadoop, HDFS, Spark, NoSQL

**Tools :** Git, Docker, Streamlit

**Visualization :** Power BI, Matplotlib, Seaborn

## EXPERIENCE

### AI & Machine Learning Intern — Bee Coders

July 2026 – August 2026

*Stack : Python, Flask, Symphony, MySQL, NLP*

- Cleaned 5 heterogeneous data sources and benchmarked 3 NLP approaches (TF-IDF, fuzzy matching, multilingual embeddings) for intent recognition
- Built a Flask API and a Symphony web application with secure authentication and multi-conversation chat

### Software Engineering Intern — Sagemcom

August 2025 – Sept. 2025

*Stack : Python, Pandas, NumPy, Matplotlib*

- Built a Python application automating Wi-Fi test analysis in an industrial environment
- Designed statistical visualizations and a process capability analysis to validate RF measurement compliance

### AI & Data Science Intern — DeepAI

Feb. 2024 – May 2024

*Stack : Python, SQL Server, ETL, Power BI*

- Built an AI pipeline (web scraping/crawling, OCR, HTML/PDF/video cleaning, Zero-Shot classification, K-means clustering) processing 193,000+ words of educational resources
- Designed a SQL Server data warehouse and Power BI dashboards analyzing 874,000 characters of content

## PROJECTS

### Hirfatuna — AI Platform for Tunisian Craft Valorization

2026

*Personal project — Stack : Python, PyTorch, ResNet-50, Qwen3.5-VL, FAISS, RAG, Streamlit*

- Built an AI platform combining computer vision and generative AI to identify Tunisian handicrafts
- Built a 3,264-image dataset and trained a ResNet-50 model reaching 90.8% accuracy
- Integrated a RAG chatbot (Qwen3.5-VL, FAISS) and deployed the application to production with Streamlit

### HDFS Small File Optimization

2026

*Research project (PFA) — Stack : Python, Hadoop, HDFS*

- Developed an HDFS small-file merging strategy based on a modified Louvain algorithm
- Improved storage efficiency by reducing NameNode memory usage by 64.7% and HDFS blocks by 97.1%

## EDUCATION

### Engineering Cycle in Computer Science and Multimedia — Software Engineering

[2024 – Present]

*Higher Institute of Multimedia Arts of Manouba (ISAMM)*

### Bachelor's Degree in Big Data and Data Analysis

[2021 – 2024]

*Higher Institute of Multimedia Arts of Manouba (ISAMM)*

## CERTIFICATIONS

- Associate Data Scientist — DataCamp (2026)
- Data Science Essentials with Python & Introduction to Data Science — Cisco Networking Academy (2026)
- Data Science & Analytics — HP Foundation (2026)
- AI Fundamentals — Cisco Networking Academy & IBM (2024)
- Deep Learning using TensorFlow — IBM (2024)